

University of Texas at El Paso (UTEP) Institutional Review Board
Informed Consent Form for Research Involving Human Subjects

Protocol Title: *Measurements of decision-making rationality*

Principal Investigator: *Alexander Friedman*

UTEP: *Biology*

Introduction

You are being asked to take part voluntarily in the research project described below, and will be compensated for your time. You are encouraged to take your time in making your decision. It is important that you read the information that describes the study. Please ask the study researcher or the study staff to explain any words or information that you do not clearly understand.

Why is this study being done?

Approximately 1000 adult individuals, 18 years and older will be enrolling in this study. Each participant will respond to a series of cost-benefit offers in the format of a survey, individualized based on the participants selected rewards and costs. On day 1 the participants will go through some basic questions like age range, sex, if experiencing any pain, level of tiredness and then they will be able to pick topics/stories of their choice from the list of stories available and each story will have 6 preference questions and 16 decision-making questions. In any additional sessions completed, participants will go through the same format of decision-making questions. These app sessions will be conducted whenever the participant chooses to complete them.

We will chart subjects' ability to perform cognitive and moral decision-making at a strategic level. We will identify the imbalanced decision-making observed in human subjects such as overvaluation of rewards, cost, or risk in decision-making. We will use a combination of neuronal network, algorithmics and analytic models to classify the psychometric functions in our subjects. We will develop a predictive model to identify the patterns of striosomal circuit activation that led to optimal cost-benefit decision-making.

What is involved in the study?

You are being asked to be in the study because we will chart subjects' performance of cost-benefit and moral decision-making at a strategic level. The behavioral part of the experiment will be conducted using an application which logs responses to a questionnaire presented to the participants in the form of stories. This survey application will be sent by a third-party site, Prolific, and therefore the survey will be completed in the absence of a proctor/administrator. On day 1 the participants will go through some basic questions like age range, sex, if experiencing any pain, level of tiredness and then they will be able to pick topics/stories of their choice from the list of stories available and each story will have 8 preference questions and 16 decision-making questions that present cost-benefit offers to the participant to evaluate. In any additional sessions completed, participants will go through the same format of decision-making questions. After data is collected, using computational modeling we will identify the imbalanced decision-making observed in human subjects such as overvaluation of rewards, cost, or risk in decision-making.

If you decide to enroll in this study, your involvement will last at minimum 1 session completing 4 stories up to as many sessions as you decide to complete. Each session is about 20-30 minutes. Each session has 3-4 stories consisting of 6 choice preference questions and 16 decision-making questions. Time and day for each session will be determined based on the subjects schedule. All participant data is anonymized.

What are the risks and discomforts of the study?

The risks associated with this research are no greater than those involved in daily activities. There are no known or anticipated risks or discomforts associated with participation.

What will happen if I am injured in this study?

The University of Texas at El Paso and its affiliates do not offer to pay for or cover the cost of medical treatment for research related illness or injury. No funds have been set aside to pay or reimburse you in the event of such injury or illness. You will not give up any of your legal rights by signing this consent form. You should report any such injury to *Dr. Alexander Friedman*, 857-233-1843 and to the UTEP Institutional Review Board (IRB) at (915-747-6590) or irb.orsp@utep.edu.

Are there benefits to taking part in this study?

You will receive monetary compensation for completing this survey. (The successful implementation of our research plan will enable the development of new tools for assessing brain, manipulating state and physiological state in the real world; and the development of new tools to classify types of decision-makers in real time. It will be transformative to conceptual understanding of decision-making, which will create a new training capacity within the U.S. military; the ability to measure optimal decision-making in healthy officers and service members. We will create new brain circuit-based cognitive neuro-biofeedback therapies that will promote strategic thinking in officer, whose decisions determine the future of our democracy in crisis situations. Our enhanced understanding of the brain circuits that support decision-making may have implication for semi-automated decision-making by supervised artificial intelligence systems.)

What are my costs?

There are no direct costs.

Will I be paid to participate in this study?

You will not be compensated for taking part in this research study.

What other options are there?

You have the option not to take part in this study. There will be no penalties involved if you choose not to take part in this study.

Choosing to withdraw or not participate will not affect your grades, class, or university standing.

Choosing to withdraw or not participate will not affect your employment, departmental, or university standing.

What if I want to withdraw, or am asked to withdraw from this study?

Taking part in this study is voluntary. You have the right to choose not to take part in this study. If you do not take part in the study, there will be no penalty or loss of benefit.

If you choose to take part, you have the right to skip any questions or stop at any time. However, we encourage you to talk to a member of the research group so that they know why you are leaving the study. If there are any new findings during the study that may affect whether you want to continue to take part, you will be told about them.

The researcher may decide to stop your participation without your permission, if he or she thinks that being in the study may cause you harm.

Who do I call if I have questions or problems?

You may ask any questions you have now. If you have questions later, you may call *Dr. Friedman at 857-233-1843 and afriedman@utep.edu*

If you have questions or concerns about your participation as a research subject, please contact the UTEP Institutional Review Board (IRB) at (915-747-6590 or irb.orsp@utep.edu).

What about confidentiality?

Your part in this study is confidential. The following procedures will be followed to keep their personal information confidential.

The results of this research study may be presented at meetings or in publications; however, your name will not be disclosed in those presentations.

Every effort will be made to keep your information confidential. Your personal information may be disclosed if required by law.

Organizations that may inspect and/or copy your research records for quality assurance and data analysis include, but are not necessarily limited to:

- Office of Human Research Protections
- UTEP Institutional Review Board

Because of the need to release information to these parties, absolute confidentiality cannot be guaranteed.

All research involved will keep detailed electronic experimental data stored in a database. These notes will have sequentially marked pages, where any experiment will be documented in detail. All notes and protocols during experiments will be collected by our software. The database will be managed by MATLAB and Postgres SQL functions. All notes will be recorded and updated daily. It will be stored on the UTEP server. Experimental setup computers and data on the server will be backed up weekly. We will create a detailed protocol description for all functions and data as routinely done for our past publications. We will use read-me files to create a supplemental material document with a description of all the functions and procedures. Data generated from this project will be administered in accordance with both University and NSF policies, including NSF data sharing policy and implementation guidance. Friedman Laboratory follows an open data philosophy. Therefore, in our previous publications and this proposal, all data and protocols without exclusion will be deposited into the Mendeley Database. All codes used to analyze and process the data will be submitted to GitHub database. We will create special scripts that generate published figures and panels from raw data and include statistical analysis. Any inventions developed under this project will be reported and managed as provided by NSF policies. The electronic version of the final, upon acceptance for publication will be deposited to the National Library of Medicine's Pubmed Central to be made publicly available no later than one year after the official date of publication. Both the PI and the University of Texas at El Paso will have responsibilities, and hence rights, concerning access to, use of, and maintenance of original research data.

Mandatory reporting

N/A

Authorization Statement

I have read each page of this paper about the study (or it was read to me). I will be given a copy of the form to keep. I know I can stop being in this study without penalty. I know that being in this study is voluntary and I choose to be in this study.

Participant's Name (printed)

Participant's Signature

Date

Signature of Person Obtaining Consent

Date